Participant Guide - Alpine Climate Data Challenge

Introduction

Welcome to the Alpine Climate Data Challenge, a hackathon dedicated to developing predictive models and visualization solutions to monitor and forecast the effects of climate change in the Maurienne and Val di Susa valleys.

This guide provides strategies, concrete examples, and methodologies to help teams structure their work, leverage available data, and develop effective and innovative solutions.

Challenge Objectives

🎯 **Understanding and Predicting the Future Climate of the Alpine Valleys**
The main goal of the challenge is to analyze open-source climate datasets and build predictive models to anticipate future climate trends. The resulting tools will support decision-making for local authorities, businesses, and citizens.

📌 **Expected Outputs:**
✅ **Local climate models:** Simulations up to 2050 based on IPCC scenarios.
✅ **Interactive dashboard:** A tool for exploring climate data.
✅ **App prototype:** An educational solution to raise awareness in local communities.
✅ **Advanced UI/UX interfaces:** Thematic maps and interactive narratives for clear communication.

🔎 **Data Science-Based Approach**
Teams will utilize a mix of data science, machine learning, and data visualization techniques, leveraging tools like **Palantir Foundry** to manage datasets and build predictive models.

Key Insights on Future Climate Impacts

Teams must analyze concrete data to develop climate forecasts and simulations. Below are key climate phenomena to study:

🌡️ 1. Rising Temperatures and Heatwaves

- **Effects:** More days above 35°C, increased energy demand, public health risks.
- **Example:** Turin may experience a **+3°C temperature rise** by 2040, increasing blackout risks due to grid overload.

🏔️ 2. Declining Snow and Ice

- **Effects:** Reduced water resources, impacts on winter tourism, lower hydroelectric capacity.
- **Example:** Chamonix could lose **40% of its ski slopes** by 2040.

🌊 3. Increased Floods and Landslides

- **Effects:** Greater hydrogeological risks, damage to infrastructure and transportation.
- **Example:** In 2022, a landslide blocked the **Turin-Lyon railway** for three weeks.

🐦 4. Disappearance of Alpine Species

- **Effects:** Ecosystem disruptions, biodiversity loss.
- **Example: 50% of Alpine butterfly species** could vanish by 2100.

💧 5. Water Stress and Droughts

- **Effects:** Reduced water reserves for agriculture, human consumption, and hydroelectric power.
- **Example:** The **Po River** reached its lowest level in **70 years** in 2022.

🔥 6. Increase in Wildfires

- **Effects:** Forest loss, higher CO_2 emissions.
- **Example:** Wildfires in **France and Italy** have increased by **400%** compared to the previous decade.

⚡ 7. Energy Sector Impacts

- **Effects:** Lower hydroelectric output, rising energy costs, summer blackouts.
- **Example:** In 2023, **France reduced nuclear power production by 20%** due to low river water levels.

🚧 8. Damage to Transport Infrastructure

- **Effects:** Rail and road disruptions, need for resilient infrastructure investments.
- **Example:** Extreme heat deformed **rail tracks in Switzerland**.

🏕️ 9. Changes in Alpine Tourism

- **Effects:** Decline in winter tourism, need for economic diversification.
- **Example:** Ski resorts below **1,500m** may become unsustainable by 2050.

🍇 10. Changes in Agriculture and Food Production

- **Effects:** Shifts in planting and harvesting cycles, reduced wine quality.
- **Example:** The **Piedmont grape harvest** now occurs **2-3 weeks earlier** than 30 years ago.

🔍 Insight #1 - Rising Temperatures and Heatwaves

🔥 **Problem:** Hotter summers and their impact on people and the environment

Temperatures in Alpine regions are rising at nearly **twice the global average** (Gobiet et al., 2020, *Environmental Research Letters*). This phenomenon is leading to:
✅ **More heatwaves:** Temperatures exceeding 35°C in historically cool areas.
✅ **Increased water stress:** Higher evaporation rates and reduced water resources.
✅ **Impact on biodiversity:** Alpine species struggle to adapt and migrate to higher altitudes.

📌 **What to do during the Challenge?**

- Analyze historical temperature series (NOAA, Copernicus, ARPA Piemonte, Météo France).
- Develop a **temperature prediction model** up to 2050 using **LSTM or ARIMA**.
- Create an **interactive map** of the most vulnerable areas (dashboard with **Leaflet or Mapbox**).
- Simulate the impact of **heatwaves on economic activities** (winter tourism, agriculture).

🖥️ **Case Study:** The *Urban Heat Island Mapping* project revealed that **Turin will experience a +3°C temperature rise** by 2040 in urban areas, increasing the risk of **power blackouts** due to air conditioning overloads.

🔍 Insight #2 - Declining Snowfall and Glacier Retreat

🏔️ **Problem:** Impact on winter tourism and water availability

By **2050, snow cover in the Alps could decrease by 30-50%** due to global warming (Jacob et al., 2018, *Nature Climate Change*). This affects:
✅ **Water resources:** Less snow = less water available for agriculture and human consumption.
✅ **Winter tourism:** Fewer skiable snow days = loss of millions of euros for ski resorts.
✅ **Increased hydrogeological risk:** Faster glacier melt raises the risk of landslides.

📌 **What to do during the Challenge?**

- Use **Copernicus and NOAA data** to analyze **30-year snowfall trends**.
- Create a **predictive model with Random Forest or XGBoost** to estimate future snow cover.
- Map **water risk areas using geospatial analysis** (GIS, GeoPandas).
- Simulate **economic scenarios for ski resorts** and propose adaptation strategies (**artificial snowmaking, summer tourism**).

🖥️ **Case Study:** A *Météo France* study found that **Chamonix will lose 40% of its ski slopes** by 2040, forcing many resorts to either **close or shift to summer tourism**.

🔍 Insight #3 - Increased Extreme Weather Events: Floods and Landslides

🌊 **Problem:** More intense rainfall and hydrogeological risks

The Alps are experiencing **more frequent extreme weather events** like cloudbursts, landslides, and mudflows. According to the **IPCC**, extreme precipitation will **increase by 15-25% by 2050**.
✅ **More floods:** Soils unable to absorb water = **urban and rural flooding**.
✅ **More landslides:** Permafrost thaw destabilizes mountain slopes, increasing **rockfalls**.
✅ **Infrastructure damage:** Railways, roads, and tunnels are **more vulnerable** to interruptions and costly repairs.

📌 **What to do during the Challenge?**

- Analyze **Météo France and ARPA Piemonte data** to identify **periods of extreme rainfall**.
- Build a **landslide risk map** using **geological data and GIS models**.
- Simulate the **economic and social impact of extreme events** on key infrastructure (e.g., railway tunnels).
- Propose **mitigation solutions:** flood retention basins, smart urban drainage systems.

🖥️ **Case Study:** In **2022, a landslide blocked the Turin-Lyon railway** for 3 weeks, causing **millions of euros in losses**. A **machine-learning-based early warning system** could help prevent such events.

🔍 Insight #4 - Changes in Alpine Biodiversity

🐦 **Problem:** Loss of Alpine species and ecosystem disruption

Climate change is altering Alpine ecosystems, forcing many species to migrate **to higher altitudes** or face extinction.
✅ **Reduction of Alpine grasslands:** Vegetation zones shifting **500+ meters** upward.
✅ **Decline of iconic species: Ermine and rock ptarmigan** are losing **30% of their habitats**.
✅ **Higher wildfire risk: Hotter temperatures and drier snow** increase summer fire risks.

📌 **What to do during the Challenge?**

- Analyze **Copernicus Land Monitoring Service data** to track **vegetation shifts**.
- Develop a **predictive model on habitat loss** for vulnerable Alpine species.
- Propose **conservation strategies**, such as **natural climate refuges or ecological corridors**.

🖥️ **Case Study:** A **University of Innsbruck** study predicts that **50% of Alpine butterfly species** could disappear **by 2100** without intervention.

🔍 Insight #5 - Impacts on Energy and Transportation Sectors

⚡ **Problem:** Increased blackouts and transport network instability

✅ **Hydropower in crisis:** With less snow and glacier melt, **hydropower production in the Alps could drop by 30% by 2050**.
✅ **Summer blackout risks:** Higher **energy demand** for air conditioning + **lower hydroelectric output**.
✅ **Railway risks:** Extreme heat **warps train tracks**, slowing transport.

📌 **What to do during the Challenge?**

- Simulate **hydropower production loss** using past climate data and future projections.
- Analyze **heatwave impacts on railway networks**.
- Propose solutions like **smart electrical grids, diversified energy sources, and rail infrastructure adaptations**.

🖥️ **Case Study:** In **2023, record-breaking heat forced France to cut nuclear power production by 20%** due to low river levels, reducing cooling capacity for reactors.

📌 Even if you've never worked with this data before, no problem!

This challenge is an opportunity to explore, learn, and experiment with complex datasets in a collaborative and supportive environment.

Available Data and Useful Sources

To develop the required solutions, you will have access to several sources of public climate data:

- **ARPA Piemonte:** Regional meteorological monitoring.
- **Info Climat:** Real-time meteorological observations.
- **Météo France:** Climate data and forecasts.
- **NOAA:** Historical climate conditions data.
- **Copernicus Climate Data Store:** Global and regional climate and environmental data.
- **ISPRA:** Italian environmental indicators.
- **DREAL Auvergne-Rhône-Alpes:** Access to regional environmental data.

📌 **Tip:** For a more in-depth analysis, compare **local datasets with global ones** and validate forecasts with observed data.

🔬 What is Climate, and Why is It Important to Study Climate Data? 🌍

Climate is much more than just the weather forecast! While weather changes from day to day, climate represents the **long-term average of atmospheric conditions** in a specific area over decades or centuries. Studying climate means looking at the past to predict the future, analyzing how **temperatures, precipitation, winds, and other climatic factors** evolve over time.

Why is it crucial to monitor climate data? 🌡️ 📊

✅ **Predicting extreme events:** Heatwaves, droughts, and floods are becoming more frequent and intense.
✅ **Anticipating impacts on communities and the environment:** From **hydrogeological risks** to **food security**, every sector is affected by climate.
✅ **Planning for the future:** Climate data help governments, businesses, and citizens make informed decisions about **urban planning, agriculture, energy, and more**.

📊 Types of Climate Data: Where Do the Information Come From?

To study climate, scientists use a combination of **historical, real-time, and projected data**, each playing a fundamental role:

🔹 Historical Data 📜

✔ Collected from **weather stations, satellites, and climate archives**.
✔ Help analyze long-term trends, such as **global temperature rise**.

🔹 Real-time Data 🌐

✔ Provided by **atmospheric sensors, ocean buoys, and meteorological satellites**.
✔ Used for **immediate monitoring** and to update forecasting models.

🔹 Projection Scenarios 🚀

✔ Based on **mathematical models that simulate future climate** under different greenhouse gas emission scenarios.
✔ Allow scientists to **assess the consequences of climate policies** (e.g., **CO_2 reduction vs. inaction**).

📌 **Case Study:** The **ERA5 dataset** from Copernicus enables the **comparison of temperature trends over the past 50 years**, highlighting the warming trend in the Alps.

🧠 Climate Models: How Do We Predict the Future Climate?

A **climate model** is a **numerical simulation** that replicates the behavior of the Earth's climate system using **mathematical equations** to describe **atmospheric currents, oceanic movements, and biosphere interactions**.

Main Types of Climate Models Used by Scientists:

🔹 Empirical Models 🏛️

✔ Based on **statistical correlations derived from historical data**.
✔ Useful for identifying **trends and anomalies** but less reliable for long-term predictions.

🔹 Dynamic Models 🔬

✔ Describe climate through **complex physical equations** that simulate atmospheric and oceanic behavior.
✔ Used for **advanced meteorological and climate simulations**.

🔹 Global Circulation Models (GCMs) 🌍

✔ Simulate the **entire Earth's climate system** on a large scale.
✔ Fundamental for the **climate projections of the IPCC (Intergovernmental Panel on Climate Change)**.

🔹 Regional Climate Models (RCMs) 🏔️

✔ Provide **detailed simulations** for **specific areas**, such as the **Alps**.
✔ Allow **local impact assessments** of climate change.

📌 **Example of Application:**
✅ **Analyzing past temperatures with an ARIMA model** can highlight **anomalies and significant trends over time**.
✅ **For future climate projections in the Alpine valleys,** IPCC data can be integrated with **advanced simulations using WRF (Weather Research & Forecasting Model)** to **assess impacts on precipitation and snow cover up to 2050**.

🚀 **The challenge is on:** Can you develop models that accurately predict future climate impacts? 🌍🔬

Comprehensive Analysis Strategy for the Challenge

This guide provides **a detailed overview of analysis strategies** to achieve all the required outputs:
✅ **Local climate models**
✅ **Interactive dashboards**
✅ **Educational app prototype**
✅ **Advanced UI/UX components**

📊 Analysis Strategies for the Alpine Climate Data Challenge

1️⃣ Data Cleaning and Preprocessing

📌 **Objective:** Ensure data quality, consistency, and completeness to achieve accurate results in climate models and visualizations.

📌 Key Steps

✅ 1.1 Data Quality Check

🔹 **Objective:** Remove errors, inconsistencies, and missing values to ensure analysis reliability.
🔹 **Techniques:**

- **Identifying missing and anomalous values** → Apply data imputation techniques (e.g., moving average, linear interpolation, regression).
- **Removing duplicates** → Identify and eliminate duplicate entries in datasets.
- **Detecting climate outliers** → Use statistical methods (e.g., Z-score, IQR, Grubbs' test).

📌 **Application to outputs:**
✅ **Climate Models:** Clean data ensures more accurate predictions.
✅ **Interactive Dashboard:** Removes errors that could distort visualization.
✅ **Educational App:** Clean data prevents errors in quizzes and interactive interfaces.

✅ 1.2 Normalization and Standardization

🔹 **Objective:** Standardize measurement units for consistent variable comparisons.
🔹 **Techniques:**

- **Min-Max normalization** for different scales (e.g., temperature in °C, precipitation in mm).
- **Z-score standardization** for normally distributed data.
- **Log transformation** to normalize skewed data distributions.

📌 **Application to outputs:**
✅ **Climate Models:** Improves generalization in machine learning models.
✅ **Dashboard & UI/UX:** Ensures consistent and readable visualizations.

✅ 1.3 Interpolating Missing Data

🔹 **Objective:** Estimate missing values to avoid errors in predictive models.
🔹 **Techniques:**

- **Geostatistical Kriging** → Useful for spatial data (e.g., temperature distribution in the Alps).
- **Cubic Spline** → Suitable for long-term climate trend interpolation.
- **Linear Temporal Interpolation** → Fills gaps in time-series datasets.

📌 **Application to outputs:**
✅ **Climate Models:** Interpolated data improves predictive accuracy.
✅ **Interactive Dashboards & Maps:** Reduces data discontinuities in visualizations.

✅ 1.4 Temporal Aggregation

🔹 **Objective:** Ensure consistency in climate data across uniform time scales (daily, monthly, seasonal).
🔹 **Techniques:**

- **Data resampling** → Convert daily data into monthly/seasonal averages for long-term trend analysis.
- **Moving window smoothing** → Reduce variability using moving averages.

📌 **Application to outputs:**
✅ **Climate Models:** Reduces noise for more stable simulations.
✅ **Interactive Dashboard:** Enables climate scenario comparisons at different time scales.

✅ 1.5 Noise Reduction in Data

🔹 **Objective:** Eliminate random variations to extract meaningful trends.
🔹 **Techniques:**

- **Smoothing filters** (e.g., Kalman, Savitzky-Golay) to reduce climate data variability.
- **Moving Average & Median Filtering** to remove erratic daily fluctuations.

📌 **Application to outputs:**
✅ **Climate Models:** Improves predictive model stability.
✅ **Interactive Dashboard & UI/UX:** Provides clearer and more consistent graphs.

2️⃣ Identifying Climate Patterns and Changes

📌 **Objective:** Extract meaningful trends from climate data to build predictive models, visualizations, and educational tools.

📌 Key Steps

✅ 2.1 Correlation Analysis

🔹 **Objective:** Identify relationships between key climate variables.
🔹 **Techniques:**

- **Correlation heatmaps** → Display interactions between temperature, humidity, pressure, etc.
- **Multiple regression** → Determine the influence of multiple variables on a phenomenon.

📌 **Application to outputs:**
✅ **Climate Models:** Helps select the most relevant features for prediction.
✅ **Dashboard:** Enables visual representation of correlations between climate variables.

✅ 2.2 Time Series Analysis

🔹 **Objective:** Extract trends and periodicity from climate data.
🔹 **Techniques:**

- **STL Decomposition** → Breaks a time series into trend, seasonality, and residual components.
- **Fourier Transform & Wavelet Analysis** → Identifies cyclic patterns in data.
- **ARIMA & LSTM** → Forecasts based on historical data.

📌 **Application to outputs:**
✅ **Climate Models:** Predicts future temperature and precipitation trends.
✅ **Interactive Dashboard:** Displays historical climate trends and future projections.

✅ 2.3 Extreme Event Detection

🔹 **Objective:** Identify climate anomalies such as heatwaves, droughts, and floods.
🔹 **Techniques:**

- **Clustering (K-means, DBSCAN)** → Groups similar climate events.
- **Statistical thresholds (percentiles, standard deviations)** → Detects outliers.

📌 **Application to outputs:**
✅ **Climate Models:** Essential for simulating future extreme events.
✅ **Dashboard:** Allows users to explore the frequency and distribution of extreme events.

✅ 2.4 Spatial Mapping & GIS

🔹 **Objective:** Represent the spatial distribution of climate variables.
🔹 **Techniques:**

- **GeoPandas, Leaflet, Mapbox** → Create interactive climate maps.
- **Spatial interpolation** → Kriging to estimate climate variables on geographic grids.
- **GIS Overlay** → Superimpose different layers (e.g., temperature, precipitation, infrastructure).

📌 **Application to outputs:**
✅ **Interactive Dashboard:** Displays future climate scenarios on geospatial maps.
✅ **UI/UX:** Enables intuitive and detailed visual storytelling.

📌 Conclusion

🔹 **Data Cleaning & Preprocessing** → Ensures accuracy and consistency in climate data.
🔹 **Pattern Identification** → Extracts insights to build effective predictive models.
🔹 **Spatial Mapping & UI/UX** → Transforms complex data into interactive and intuitive tools.

📌 **Expected Outcome:**
An **optimized data pipeline** that supports **robust predictive models, interactive dashboards, and educational tools** to address the climate challenge in the Alps. 🚀

💻 Infrastructure and Development Environments

To facilitate climate data analysis and predictive model development, participants will have access to **SoBigData and Palantir Foundry**, two powerful platforms offering **advanced computing and visualization capabilities**.

✅ **SoBigData** → Dedicated environment for **model processing and development**, with **GPU access and high-performance computing** resources to optimize model training.
✅ **Palantir Foundry** → Recommended platform for **interactive dashboards, advanced visualizations, and app prototyping**, but also usable for **data processing and model development**.

🚀 **Tool Flexibility**
Teams are free to:
🔹 **Follow the suggested workflow:** SoBigData for modeling, Foundry for visualization.
🔹 **Use only SoBigData** for the entire process.
🔹 **Use only Foundry,** integrating data processing tools.
🔹 **Integrate other environments** (Google Colab, Jupyter, AWS, etc.), ensuring that the final model can run on SoBigData.

📌 **Important!**
❌ **Pre-trained models are NOT allowed:** All models must be developed **from scratch** using the provided or self-sourced climate data.

🔹 If developing the model in external environments, ensure that the **code and model** are **compatible and executable** in SoBigData.

This means:
✅ **Scalability:** The model should not depend on hardware resources unavailable in SoBigData.
✅ **Optimized code:** The training process must run efficiently within SoBigData.
✅ **Compatible libraries:** Use supported libraries to prevent portability issues.

📌 **Presentation**
🚀 Only finalist teams will proceed to the **pitch competition**, where they must develop a structured presentation to showcase their results to the jury.

📌 **Practical Tip:**
Test your model on **SoBigData** from the early stages to avoid last-minute adjustments. **Working in a consistent development environment** will reduce errors and improve project stability.

🚀 **Now it's your turn:** What strategy will you adopt to optimize your development pipeline? 💡

Required Outputs

1️⃣ Local Climate Models

📌 **Objective:** Create climate simulations up to 2050 based on IPCC scenarios for the **Maurienne and Val di Susa valleys**.

📌 **Required Deliverables:**

✅ Code and Mathematical Model

- Python/R scripts for climate modeling (**LSTM, ARIMA, Random Forest, GCM, WRF**).
- Use of **open-source climate datasets** (Copernicus, NOAA, ARPA Piemonte, ERA5).
- Integration with **local historical data** and **future simulations**.
- **Model validation** using observed data.

✅ Technical Report (Executive Summary + Model Documentation)

- **Executive Summary (1-2 pages) including:**
  - Model objectives.
  - Data used.
  - Methodology applied.
  - Key insights obtained.
- **Technical Documentation including:**
  - Model explanation (**equations, neural network architecture, data preprocessing**).
  - **Results analysis** with graphs and tables.
  - **Comparison between future scenarios and historical climate trends**.

✅ Final Output

- [ ] **CSV/JSON files** with simulated climate data for different scenarios.
- [ ] **Visual report** with **graphs** on climate projections (**temperature variations, precipitation, extreme events**).
- [ ] **Executive Summary**.

2️⃣ Interactive Dashboard

📌 **Objective:** Develop an interface to explore climate data and make simulations accessible.

📌 **Required Deliverables:**

✅ Dataset

- **Datasets used**.
- **Structured datasets** with output from climate models.

✅ Interactive Frontend (Web/App Dashboard)

- **Visualization panels** including:
  - **Dynamic charts** on temperature, precipitation, snow, wildfires, etc.
  - **Interactive geospatial maps**.
  - **Scenario selectors** (e.g., *"High Emissions," "Moderate Emissions," "Low Emissions"*).

- **Exploration tools** including:
    - **Filters for geographic area and forecast period**.
    - **Comparison between historical and future data**.

✅ Dashboard Documentation

- ☐ **Short user guide** (PDF or web page).
- ☐ **Architectural schema** (e.g., data flow, APIs, databases).
- ☐ **Demo video of the interface**.

3️⃣ Educational App Prototype

📌 **Objective:** Create an app to raise awareness among **students, institutions, and citizens** about climate change impacts in the **Alps**.

📌 **Required Deliverables:**

✅ App Structure

- **Interactive storytelling:** Narrative content + climate simulations.
- **Quizzes and gamification:** To engage users.
- **Animated maps:** Show climate evolution over time.

✅ Suggested Technologies

- **Mobile/web frameworks:** Flutter, React Native, Progressive Web App.
- **Open Data integration:** For real-time climate data updates.

✅ Required Output

- ☐ **App wireframe** (main screens).
- ☐ **Working prototype or interactive demo**.
- ☐ **Developer documentation**.
- ☐ **Project pitch presentation (video/slides)**.

4️⃣ Advanced UI/UX Components

📌 **Objective:** Develop **thematic maps** and **interactive narratives** to effectively communicate climate data.

📌 **Required Deliverables:**

✅ Advanced UX/UI Design

- **Clear and intuitive interface**, optimized for **desktop and mobile**.
- **Accessible colors and icons** (considering color blindness and other needs).

✅ Climate Data Visualization

- **GIS maps with climate overlays** (e.g., **temperature levels, snow cover, wildfires**).
- **Dynamic infographics with animations**.
- **Interactive timelines** to visualize past and future climate changes.

✅ UI/UX Documentation

- [ ] **Report on design logic used**.
- [ ] **Usability analysis**.

Conclusion

The **Alpine Climate Data Challenge** is a unique opportunity to test your skills in **data science and artificial intelligence**.

Make the most of the available data, **experiment with innovative techniques**, and most importantly, **collaborate effectively with your team**!

🚀 **Good luck!**